# Unified Funeral Services Management Platform

Data Flow Diagram

## Data Sources

- First Call Forms
- Handwritten Vitals Forms
- Family Portal
- Third-Party Systems
- IoT / OBD-II Telemetry

## Data Ingestion Layer · FSD-15

- Form Submission Pipeline
- OCR / AI Digitization — FSD-09
- API Ingestion Endpoints
- Batch / Scheduled Imports
- Stream Processing — Kafka

## Data Validation & Quality · FSD-16

- Schema Validation
- Business Rule Engine
- Anomaly Detection
- Duplicate Detection
- Completeness Checking

## Data Governance & Stewardship · FSD-17

- Data Dictionary — FSD-13
- Metadata Repository — FSD-18
- Data Lineage Tracking
- RBAC Enforcement — FSD-12
- Audit Logging — FSD-27

## Core Data Storage

- **Operational DB (OLTP)** — Cases · Members · Payments · Inventory
- **Data Warehouse (OLAP)** — Historical data · Aggregates · KPIs
- **Document Store (Blob/S3)** — Scanned docs · Photos · Contracts
- **Analytics Data Lake** — Raw · Processed · ML features · Models
- **Audit & Log Store** — All actions · Access logs (immutable)

## Analytics & Insights · FSD-21

- Executive Dashboards — FSD-22
- Operational Reports — FSD-22
- Predictive Models — FSD-30
- Embedded BI — FSD-21
- Self-Service Analytics — FSD-21

## Data Consumers & Outputs

- Family Portal — FSD-07
- Funeral Directors
- Regulatory Reporting — FSD-23
- Partner Systems — FSD-25
- External APIs — FSD-31